

		$M_B = 1.96 \times 10^{30} = 2.0 \times 10^{30} \text{ kg}$ $M_A = 7.8 \text{ or } 8.0 \times 10^{30} \text{ kg}$	1
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3	Ice is less dense than liquid water due to the structure of the bonds between the molecules when it is in a solid state. Hence, when ice melts into water at 0°C, the density of water increases. As the liquid water increases in temperature, its density further increases to a maximum value at 4.0°C.		
	(a)	Consider a 5.0 kg block of ice at 0°C that is melted into liquid water at the same temperature using a heat source supplying a constant power of 4180 W.	
	(i)	The specific latent heat of fusion of ice is $3.34 \times 10^5 \text{ J kg}^{-1}$. Determine the time taken to completely melt the block of ice into liquid water at 0°C. time = s [3]	
		Solution: The heat required to melt the 5.0 kg block of ice, Q: = mass $\times$ latent heat of fusion	

Use the words **positive**, **negative** and **zero** to complete Table 3.1 for the three terms in the equation for each of the processes shown. You may use each word once, more than once, or not at all.

Table 3.1

Process	ΔU	q	w
Ice melting into liquid water at 0°C			
Liquid water warming up from 0°C to 4°C			

[2]

[Total: 8]

Solution:

As the ice melts, its internal energy increases due to the energy used to increase the intermolecular separation. The source of this energy is the heat supplied to the ice and the work done on the ice by the atmospheric pressure (since the ice contracts as it melts)

As the liquid water increases in temperature, its internal energy increases. The source of this energy is the heat supplied to the water from the surroundings and the further contraction of the volume of the water (as its density continues to increase) means that the atmospheric pressure does further work on the water.

Process	ΔU	q	w
Ice melting into liquid water at 0°C	positive	positive	positive
Liquid water warming up from 0°C to 4°C	positive	positive	positive